

Mason Scott

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EDUCATION

University of Maryland

Bachelor of Science in Computer Science – Data Science Track, Minor in Statistics

College Park, MD

May 2026

EXPERIENCE

Experio AI

AI Engineering Intern

Bethesda, MD

June 2025 – November 2025

- Built structured data extraction and normalization pipelines to parse large, unstandardized RFP documents into clean, retrieval-ready inputs for a retrieval-augmented generation (RAG) system.
- Designed multi-stage agent workflows (planning, extraction, validation, retrieval, refinement) that queried the company's Neo4j knowledge graph to ground LLM outputs in structured proposal data.
- Independently researched LangGraph and CrewAI, developing solo prototype agents in a separate repository to evaluate design tradeoffs before production integration.
- Built and tested a client-facing proposal generation agent demo that integrated the Neo4j-backed retrieval system with a proposal generation pipeline, authoring comprehensive tests in a shared production codebase.

High Sierra Pools

Pool Manager

Frederick, MD

May 2024 – Aug. 2024

- Led and supervised a team of lifeguards, organized daily operations, and coordinated emergency responses with EMS/police.

PROJECTS

EngageCS – Student Engagement Analytics Platform | *Python, Dash, Flask, PostgreSQL, Plotly, Docker, Valkey*

- 6-person capstone commissioned by a UMD CS instructor; deployed to production and used live in lectures with ~140 students per session, Spring 2026
- Data engineer and analytics lead; designed the PostgreSQL schema, built the Canvas LMS sync pipeline (courses, sections, rosters), and led a V2 schema migration that unblocked all Canvas-dependent acceptance tests
- Built two internal Python libraries: a data processing lib (attendance rates, risk scores, participation metrics, week-over-week deltas) and a registry-driven Plotly viz lib with a clean public API, decoupling computation from presentation
- Delivered instructor, TA, and student dashboards with risk badges and topic engagement analytics; system grew 31% beyond original scope with all core acceptance tests signed off after live classroom validation

Impulse – Full Stack Music Recommendation App | *React, FastAPI, Docker, Redis, Celery, PostgreSQL*

- Built a full-stack music recommendation system with containerized microservices using Docker Compose
- Designed and deployed a FastAPI recommendation engine connected to Redis and Celery for async background task processing and song featurization pipelines
- Built frontend and backend layers to consume Spotify API data, serve user recommendations, and manage OAuth2 auth

Air Quality Predictor | *Python, Streamlit, TensorFlow, AWS, MySQL, OpenWeatherMap API, Plotly*

- Developed and deployed an LSTM-based air quality prediction model using TensorFlow, achieving a test RMSE of 0.1668, integrated with real-time data from the OpenWeatherMap API
- Built a scalable data pipeline using MySQL on AWS and a Streamlit web app for real-time air quality monitoring and predictions

TECHNICAL SKILLS

Languages: Python, C, Java, SQL, R, Rust, SAS

Web: JavaScript, HTML/CSS

AI / Agentic Systems: LangChain, RAG, NLP, Neo4j

Frameworks & ML: React, Dash, FastAPI, Flask, TensorFlow, Streamlit

Tools / Data: Git, Docker, PostgreSQL, AWS, Redis/Valkey, NumPy, Pandas, Scikit-learn, Plotly, Matplotlib